

Feature Extraction in Retinal Fundus Images

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Abstract—An automated method for Retinal feature extraction is done for the diagnosis of several retinal diseases such as Diabetic Retinopathy (DR). In this paper, a new adaptive histogram approach is discussed to extract the features. Detection of Optic disk (OD) forms an important step in the automatic screening process for risk population at very low cost. OD extraction is quite difficult, since its brightness, color and contrast of OD is similar to other components such as cotton wools & exudates. OD serves as the landmark for fundus feature, namely fovea, macula and blood vessels, have predefined constant distance from OD center. The proposed method includes binary thresholding of fundus images followed by morphological operation to remove small objects. Then, the boundary of OD is traced using boundary tracing technique. The method is tested in the fundus images, from the screening section of Mahatma Gandhi Eye hospital & Research center, Tamilnadu. The centroid, perimeter and area for detected OD is evaluated & tabulated for further diagnosis. The evaluated value is appreciable and comparable to the standard methods available in literature.

Keywords—Diabetic Retinopathy, Feature, Optic disk, binary thresholding, morphological structuring element.

I. INTRODUCTION

Ocular fundus images provide necessary information about retinal and various systemic and non-systemic diseases. Fundus image analysis is one of the non-invasive techniques, and needs automatic method to avoid traditional manual grading. Diabetic Retinopathy (DR) is one of the leading ocular diseases. The WHO predicts that 135million people around worldwide are having diabetes and this will rise to 300 million by 2025. The statistics of occurrence will raise double by 2050. The persons having diabetes is affected by DR, around 24,000 every year.

Screening programs for Diabetic Retinopathy requires automated or computer aided analysis of retinal images. The normal features in color fundus images include Optic disk, fovea, macula and blood vessels. The main abnormal features are microaneurysms, exudates, cotton wool spots, hemorrhages and neovascularisation.

Automated method requires accurate localization and detection of anatomical features for various diseases. The important or key features for DR and Glaucoma are Optic disk (OD) or Optic nerve head (ONH). The OD appears towards the

left side or right side of the fundus image. It is more or less circular in area, roughly about one sixth of the width of the image in diameter. OD helps to locate other landmarks in fundus images namely fovea and macula which have predefined distance from OD center. OD appears in color fundus images as a bright yellowish or white region. Its shape is interrupted by outgoing vessels.

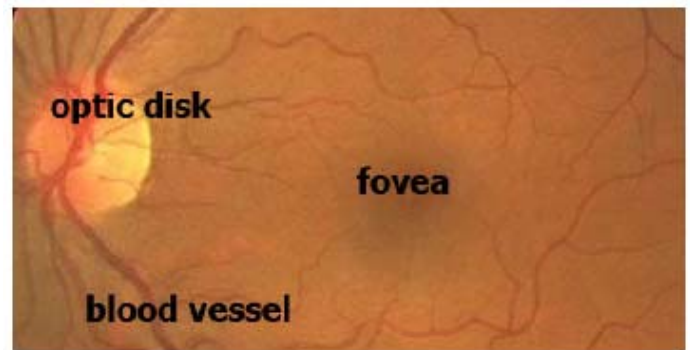


Fig. 1. Colour fundus image showing main features of retina

Optic disk is the brightest part in the normal fundus images. OD can be seen as a pale, round or oval disk [1]. The brightest pixels are clustered & the OD is located by the centroid of the largest cluster [2][3]. In 4(b) [4], disk is identified by the area with the highest variation in intensity of adjacent pixels. The geometrical relationship between OD & bloodvessels are utilized, which is a top-down method. In [5][6], OD is the points of retinal nerve fibers from the eye and the entrance and exit point for retinal blood vessels. Any change in shape, depth or color in or within OD is used by ophthalmologists to assess various retinal pathologies. [7]. It also serves as a reference to measure distance and identify other anatomical parts in retinal images. OD segmentation is a challenging task due to blood vessel occlusions, ill-defined boundaries, image variations near disk boundaries due to diseased condition and variable imaging conditions. [8] Diameter lies between 40 to 60 pixels in 640x480 color photographs.

The paper is organized as follows. In section II, Related works for Optic disk segmentation are briefly mentioned. Section III

discuss about the proposed method. In Section IV Results and future scope are discussed.

II. RELATED WORKS

In [9], a new approach is proposed to automatically extract the features namely optic disk by Principal Component analysis (PCA) and shape detection by modified active shape model (ASM).

Detection of OD from green channel is developed in [8], for automatic screening process for diabetic retinopathy. Identification of abnormal signs is used in detecting more eye diseases. OD detection employed average filtering, contrast stretching, thresholding and morphological gradient & boundary tracing techniques.

S.Sekar et al. [10], described a novel method to automatically localize the important feature, the OD. The method consists of two steps: in the first step, a circular region of interest is found by first isolating the brightest area in the image by means of morphological processing. In the second step, Hough transform is used to detect the main circular feature in the positive horizontal gradient image within the region of interest.

In [7], Robust optic disk segmentation from colour retinal images is proposed. Region based active contour model that uses local image information around each point of interest in multidimensional feature space. Local energy functional and level-set representation to achieve desired OD segmentation. Local energy functional defined on each image point provides sufficient information to determine, which is robust to the variations found in and around optic disk region.

Automatic detection of the optic nerve head and localization of the macula, by using geometric model of the vasculature is discussed by Kenneth W.Tobin[11].

Amin Dehghani et.al [12], proposed a new method for localizing optic disk using histogram matching technique.

Carmona et.al , used genetic algorithm method for curve fitting the ellipse for approximation of optic disk in retinal images. A model is developed for geometrical properties and intensity levels similar to optic disk contour pixels.

Siddalinagaswamy et.al , developed a new approach for automatic localization of OD boundary. Iterative thresholding combined with connected component analysis are used.

Osareh , proposed a method based on template matching for localizing Optic disk. Templates for optic disk are created and the correlation between each image and template is computed. The point having maximum correlation value is selected as the optic disk center.

Automatic segmentation of Hard Exudates in the fundus images based on boosted soft segmentation is discussed. It developed a multi-scale background subtraction method to obtain the coarse segmentation result. Hard exudates are extracted using a SVM classifier.

III. PROPOSED METHOD

The proposed method has been tested and evaluated on two publicly available databases namely DRIVE and DIARETDB1 database. DIARETDB1 provides 89 images with a variety of diagnoses captured by a Topcon TRV-50 fundus camera at 45 deg FOV in PNG format. Drive, photographs are obtained from DR screening program in Netherlands, used for clinical diagnoses. Images acquired using Canon CR5 non-mydratic 3 CCD cameras with a 45 deg FOV, in TIFF format.

The proposed method has two main processing phases one is preprocessing which includes binary thresholding of fundus images followed by morphological operation to remove small objects. Then, second is the boundary detection of OD by using boundary tracing technique. Each of these phases is detailed and illustrated.

Input image is a RGB image and only green channel/Gray scale image is to be extracted. This channel image has a better contrast when compared with the other two channels. Better contrast is achieved between the background images with the optic disk. Hence, gray scale image is selected for preprocessing. This channel is considered as the natural basis for OD segmentation algorithms. The preprocessing step is involved in order to avoid or reduce the non-uniform illumination of images during acquisition, removal of noise present etc. and make image suitable for OD extraction.

A. Preprocessing of Retinal image

Preprocessing steps.

1) *RGB to Gray image*: First is the conversion of RGB images in to a Gray scale image using suitable command. The Optic disk appear brighter than its background. Proper resizing of image is also done.

2) *Contrast-limited Adaptive Histogram Equalization (CLAHE)*
This adaptive histogram equalization enhances the contrast of the gray scale image. CLAHE operates on small regions in the image, called *tiles*, rather than the entire image. Each tile's contrast is enhanced, so that the histogram of the output region approximately matches the histogram specified by the Distribution parameter. The neighboring tiles are then combined using bilinear interpolation to eliminate artificially induced boundaries. The contrast, especially in homogeneous areas, can be limited to avoid amplifying any noise that might be present in the image.

3) *Binary thresholding*: Converts the grayscale image to binary by thresholding. The output binary image BW has values of 1 (white) for all pixels in the input image with luminance greater than LEVEL and 0 (black) for all other pixels. Proper value of threshold is chosen for training and fixed automatically.

4) *Background Noise Elimination*:

Mathematical Morphology:

Morphological operation is done to remove the noise present in the background of retinal image. Morphological open is done to

remove the blood vessels present whose diameter is smaller compared with OD and smoothes the inner small vessels in the Optic disk region. Opening is based on the morphological operations erosion and dilation. Opening smoothes the inside of the object contour, breaks narrow strips and eliminates thin portion of the image. It is done by first applying erosion and then dilations on the given retinal image. Opening operation is used to remove noise and CCD defects.

Morphologically open binary image (remove small objects)
The opening operation can be mathematically represented as

$$X \circ B = (X \ominus B) \oplus B \quad (1)$$

X - Input image, B - Structuring element

Erosion can be represented as

$$(X \ominus B)(x, y) = \min \{X(x + s, y + t) \mid (s, t) \in B\} \quad (2)$$

Erosion shrinks the image.

Dilation can be represented as

$$(X \oplus B)(x, y) = \max \{X(x - s, y - t) \mid (s, t) \in B\} \quad (3)$$

Dilation is the expansion of the image.

B. Boundary Tracing and Detection of Optic Disk

This phase traces the border pixels along the edges of the optic disk and returns the row and column coordinates of the border pixels. This technique uses the row and column coordinates of a border pixel as the starting point for the boundary tracing. The nonzero pixels in the binary image belong to the optic disk and pixels with zero (0) values constitute the background. The obtained optic disk has a filled and the selected Starting point is on a thin part of the optic disk. Then all the pixels of the border of the optic disk are traced on all directions and superimposed on the given image. This phase, search only for object (parent and child) boundaries which can provide better performance with no holes.

The parameters centroid, area and perimeter of the Optic disk for various retinal images are calculated and tabulated.

IV. RESULTS AND DISCUSSION

The proposed method has been tested with the images available in DRIVE and DIARETDB1 data bases. This method retains the computational simplicity and segments the Optic disk from fundus image. The results are shown in Fig.3. (a,b,c) for DRIVE database and Fig.4(a,b,c) for DIARETDB1 database.

In both the datasets a-original input image, b-gray scale image, c- Segmentation of Optic disk. From the extracted optic disk the doctors can able to predict normal and abnormal cases.

The flowchart for the proposed method is shown in figure (2). The automated method also calculates the OD area, its centroid and perimeter and is tabulated. This has been both for the both the databases, which has clinical importance to differentiate various levels and stages of DR in a very accurate and robust manner, to detect at its earlier stage itself. Any of suitable optimization technique could be employed for further improvement in segmentation, analysis and classification.

The proposed algorithm for locating Optic disk for fundus image is simple and effective compared to the other methods. The proposed method is able to detect optic disk on 33 retinal images of DIARETDB1 database and 20 retinal images of DRIVE database correctly.

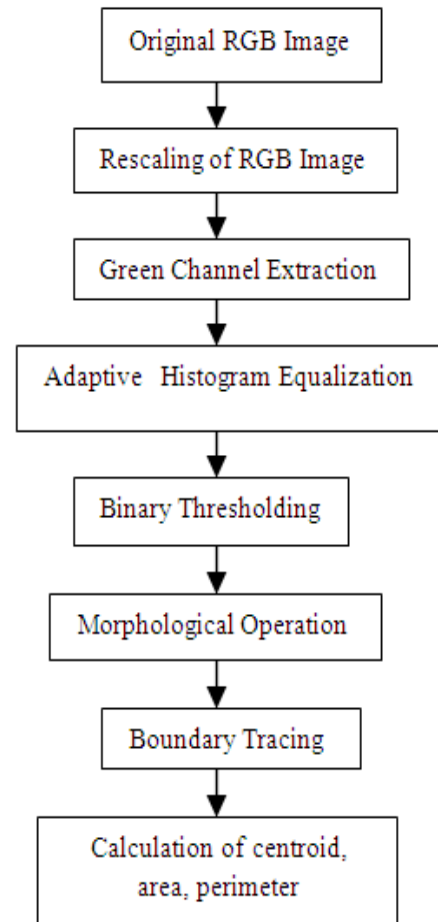


Fig.2.Flow chart: 1. Enhancement Stage



(a)

(b)

(c)

Fig.3.(a) Images for DRIVE database,(b) Gray Scale Image, (c) Optic Disk

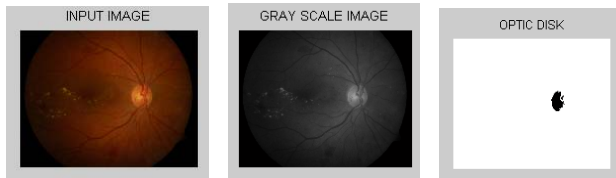


Fig.4.(a) Images for DIARETDB1,(b) Gray Scale Image, (c) Optic Disk

Tabulation

Data	Area	Centroid	Perimeter
1	4438	(583.73,279.73)	540.25
2	3478	(136.27, 259.61)	359.84
3	4049	(563.166,274.14)	404.96

Table: 1 DRIVE Database

Data	Area	Centroid	Perimeter
1	3412	(159.67,307.38)	352.43
2	3425	(477.63, 278.43)	417.34
3	3438	(491.33,268.95)	340.15

Table: 2 DIARETDB1 Database

IV.CONCLUSION

In the proposed method an adaptive histogram approach is presented to achieve OD segmentation. The scope of the method is, it traced the boundary of the optic disk by concentrating on each pixel information. The proposed method captures OD boundary in a unified manner for both normal and abnormal cases without imposing any constraint on the result. This method is able to segment the OD correctly, evident from the OD area and Centroid. Further, future improvement can be done by extracting all other features like macula and exudates, which is useful for diagnosing Diabetic Maculopathy and neovascularisation.

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